Distributed Systems

4th Lecture

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Outline

- Architectures and Algorithms
  - Matching the Architecture to the Algorithm
- Distributed Processing
  - Performance
  - Ability
  - Behavioral Principles
- Q&A

Principal Architectural Models

- Client-Server
  - Client Processes Interacting w/ Server Processes to Access Their Shared Resources Managed in a Centralized Manner

Not Scalable

    - E.g., DNS Servers, Web Search Engines

Figure 2.3 on Page 63

Principal Architectural Models (Cont'd)

- Peer-to-Peer
  - Processes Interacting Cooperatively to Access Their Shared Resources Collectively-Managed in a Distributed Fashion
    - E.g., Napster

Distributed Load, Resiliency

Figure 2.4a on Page 48

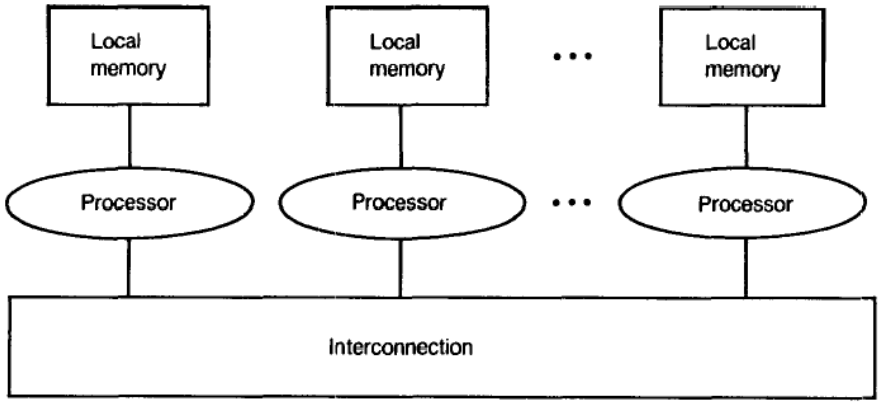
Flynn's Model for Architectures

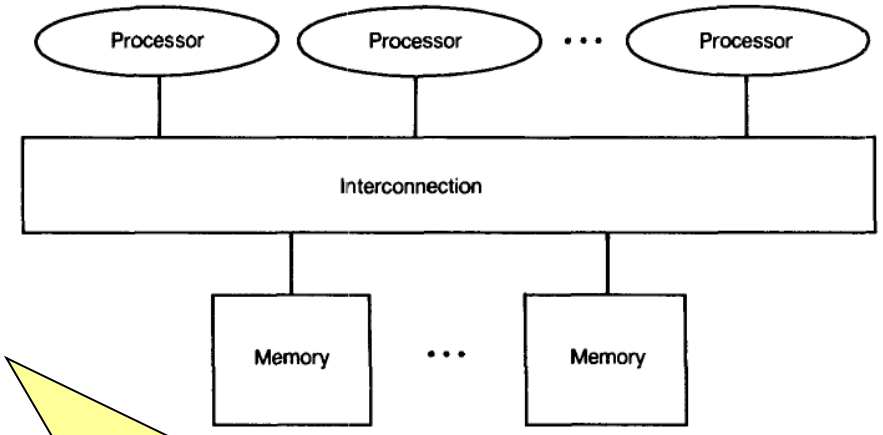
- SISD (Single Instruction Stream, Single Data Stream)
  - Uniprocessor
- SIMD (SI, Multiple Data Streams)
  - Single Instruction Memory & Control Processor w/ Memory per Processor
- MISD (Multiple Instruction Streams, SD)
- MIMD
  - Centralized Shared-Memory Architectures
  - Distributed-Memory Architectures

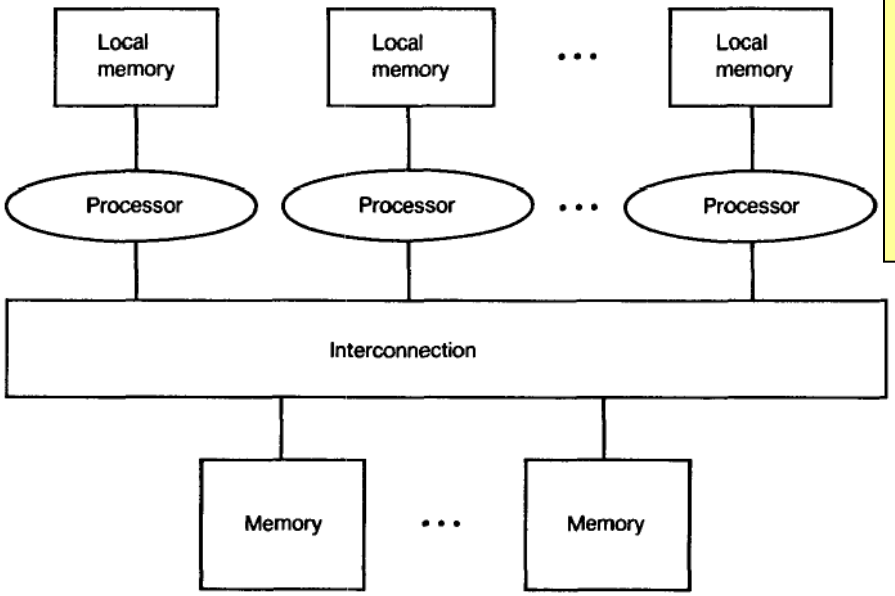
E.g., UMA, Uniform Memory Access

E.g., NUMA, Non-Uniform Memory Access

MIMD Architectures







E.g., MPP (Massively Parallel Processing) – Problem: Explicit Data Distribution

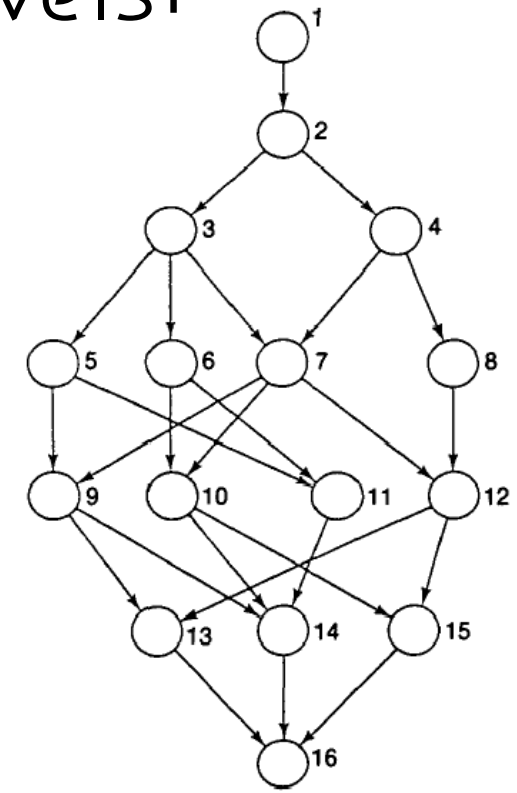
E.g., SMP (Symmetric Multiprocessing),– Problem: Poor Scalability

E.g., NUMA

Characterizing the Algorithm [Kleinrock85]

- To Exploit Its Potential for Concurrency at Different Granularity Levels:
  - Job
  - Task
  - Process
  - Instruction
  - Register Transfer
  - Logic Device

E.g., by Using a Graph Model



Matching the Architecture to the Algorithm [Kleinrock85]

- Considerations Regarding the Graph Model
  - Partitioning
    - Deciding granularity levels
    - Grouping
  - Scheduling
    - Assigning processors and memory modules
  - Memory Access
  - Interprocessor Communication
  - Synchronization
    - Preserving dependency

Matching the Architecture to the Algorithm [Kleinrock85]

- Balance and Tradeoff among Communication, Processing, and Storage
  - Trading Processing for Communication
    - E.g., data compression prior to transmission
  - Trading Storage for Processing
    - E.g., storing a list of computational results
  - Trading Storage for Communication
    - E.g., storing data from a previous communication
- Failure Detection & Recovery

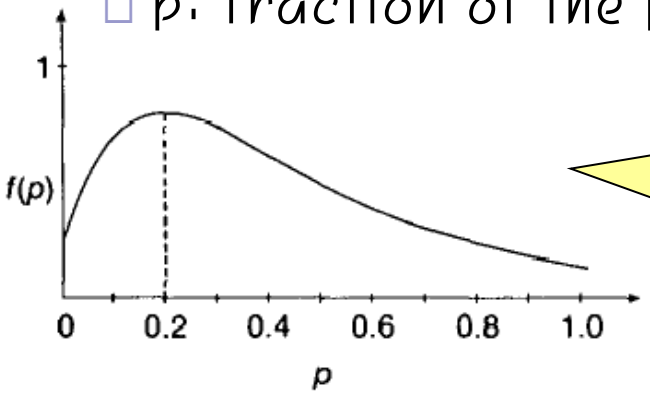
Parallel-Processing Performance

- Speedup (S)
  - Serial Computation Time T_s over Parallel Execution Time T_p ($S=T_s/T_p$)
    - $T_p = TCOMP_p + TCOMM_p$
    - $TCOMP_p = T_s(1-A)/P + T_sA$, where A is the serial portion
      - E.g., If T_s = 100, A=0.2, & P=10, then $TCOMP_p$ = 28
    - $T_p = T_s(1-A)/P + T_sA$, assuming that $TCOMM_p=0$
    - $S = T_s/T_p = T_s/(TCOMP_p + TCOMM_p) = 1/\{A+(1-A)/P\}$, assuming that $TCOMM_p=0$
      - $1 \leq S \leq P$

Amdahl Portion
– Speedup Will Be Limited by This Portion

Ability of Distributed Processing

- Possibly Producing the Best Result
  - Greedy Solutions with Unexpected Results: E.g., Goore Game [Kleinrock85] (Repeating Actions Preferentially by Trial and Error)
    - Many Players Being Asked to Vote YES or NO in Each Interval
      - Acting w/ sufficient memory in a probabilistic fashion
      - W/ each unaware of the others
    - Referee Rewarding Each Member Independently w/ a Probability Given by f(p) Unknown to It
      - p: fraction of the player set that votes YES



Exactly 20% of the Players Will Vote YES with Probability 1

We May Be Able to Explain How the Colony of Ants Perform Its Tasks

Ability of Distributed Processing (Cont'd)

- Possibly Producing a Globally Suboptimal Solution
  - Most Greedy Solutions: E.g., Prisoner's Dilemma (Predicting the Behavior of the Other Players)
    - Two Men Being Held in Separate Cells for a Crime They Did
    - District Attorney w/ Hard Evidence Just for One-Year Penalty
      - Making the following offer to each prisoner:
        - Free if confessing w/ your partner remaining silent
        - In jail for 20 years in the opposite case
        - Five years in jail for each if both confess

Dilemma: Each Is Tempted to Confess, But If Both Confess, It Will Be Worse

Behavioral Principles [Kleinrock85]

- Developing innovative architectures for parallel processing
- Providing better languages and algorithms for specification of concurrency
- More expressive models of computation
- Matching the architecture to the algorithm
- Understanding the trade-off among communication, processing, and storage
- Evaluation of the speedup factor for classes of algorithms and architectures
- Evaluation of the cost-effectiveness of distributed-processing networks
- Study of distributed algorithms in networks
- Investigation of how loosely coupled self-organizing automatons can demonstrate expedient behavior
- Development of a macroscopic theory of distributed systems
- Understanding how to average over algorithms, architectures, and topologies to provide meaningful measures of system performance